

# Integration tests and fixtures

Project: Grounded Learning Service

## 1. Your mission

### Your learning target

I can explain the mechanism and terminology of integration tests and fixtures.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Testing model-dependent behaviour. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Least privilege and tool scopes. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Typed contracts and schemas. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Explain the mechanism and terminology of integration tests and fixtures

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

### Retrieve what you already know

Recall Testing model-dependent behaviour. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

#### My starting explanation

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### This week's end goal

Explain integration tests and fixtures using the deciding step in this fictional service.

## 2. Notice the evidence

Integration tests exercise connected components and failure propagation. Fixtures supply controlled input records and mocked service outcomes.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

**List the given facts and one fact that is not supplied.**

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## 3. See how it works

Trace request validation, session permission, lookup and output validation together. Include absent sources and failing dependencies; verify that denied requests never invoke restricted lookup.

### Integration tests and fixtures — evidence trace

**Input and scope · Check or calculation · Conclusion and limit**

**Input and scope:** A student requests a teacher source; test response is denied, but a recorded lookup call read the teacher file.

**Check or calculation:** Trace request validation, session permission, lookup and output validation together. Include absent sources and failing dependencies; verify that denied requests never invoke restricted lookup.

**Conclusion and limit:** The test exposes a hidden boundary failure despite the denied final response. Assert both response and call trace; deny before lookup.

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The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

## Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

## 4. Compare the cases

### Compare a familiar case with a changed case

A student requests a teacher source; test response is denied, but a recorded lookup call read the teacher file. Show the intermediate steps and give the conclusion with its scope.

**Keep the procedure fixed. Explain what changed in the inputs and result.**

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### Change one thing in the pictured case

Change: The adapter returns M when N was requested.

Prediction: Fail the integration check: map is the wrong item.

Reason: The request now reaches evidence for another identifier.

## 5. Words that unlock the idea

### Word · Meaning

**Word:** Integration test

**Meaning:** A test of connected components

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**Word:** Fixture

**Meaning:** Controlled data or behaviour supplied to a test

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**Word:** Call trace

**Meaning:** A record of which operations occurred

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**Use two terms to explain a step in this week's example.**

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## 6. Follow a worked example

A student requests a teacher source; test response is denied, but a recorded lookup call read the teacher file.

The test exposes a hidden boundary failure despite the denied final response. Assert both response and call trace; deny before lookup.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

### A pictorial worked case: Integration tests and fixtures

Catch a wrong adapter connection. A service reads a fake catalogue fixture and answers a request for item N.

### The facts we will use

Fixture M="map", N="notebook". No external network is involved.

## Integration tests and fixtures

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Load fixture

The test supplies fixed records M and N.



#### 2. Follow identifier

The application must request N from its adapter.



#### 3. Check full reply

The returned item should be notebook with source N.

### CHANGE ONE THING

The adapter returns M when N was requested.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Load fixture

**What happens and why:** The test supplies fixed records M and N.

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**Step:** 2. Follow identifier

**What happens and why:** The application must request N from its adapter.

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**Step: 3. Check full reply**

**What happens and why:** The returned item should be notebook with source N.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

Integration testing checks how components connect. Each component can work alone while their identifier exchange is wrong.

### What this example does not establish

A controlled fixture does not prove a live catalogue behaves identically; also review live adapter contracts.

## 7. Find and repair a mistake

A mock verifies declared interactions, not real provider availability or all runtime limits.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

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### Locate the earliest unsupported step

A report omits this qualification: A mock verifies declared interactions, not real provider availability or all runtime limits. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**State one repair and the checks you would rerun.**

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## 8. Practise together

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

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**Explain your prediction to a partner before comparing results.**

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### Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Integration tests and fixtures”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

#### Plan · Do · Check

**Plan:** What is given? Which rule or procedure applies?

**Do:** Record each intermediate step.

**Check:** Does the result follow? What remains unknown?

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Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for Integration tests and fixtures. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

Follow request N through the fixture and adapter. Which source and item should appear in the reply, and what precisely fails when the adapter returns M?

#### My explanation using a pictured step and the given facts

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This is supported practice. Keep the separately named independent case for an attempt before feedback.

## 9. Run the investigation

### Predict, investigate and preserve the evidence

A student requests a teacher source; test response is denied, but a recorded lookup call read the teacher file. Show the intermediate steps and give the conclusion with its scope.

#### Record · Your evidence

**Record:** Given input and fixed rule

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Prediction before checking

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Actual result, or “not run”

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** First mismatch and one explanation

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Boundary or missing input and expected response

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

### Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

## 10. Build your understanding

### Explain → apply → challenge

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

A report omits this qualification: A mock verifies declared interactions, not real provider availability or all runtime limits. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

**Explain which evidence supports your conclusion and which claim remains unproved.**

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## 11. Connect your project

Add a labelled integration tests and fixtures evidence sheet to Grounded Learning Service. Show how you can explain the mechanism and terminology of integration tests and fixtures. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

### Project evidence · Record

**Project evidence:** Component and version

**Record:** \_\_\_\_\_

\_\_\_\_\_

**Project evidence:** Inputs permitted at prediction time

**Record:** \_\_\_\_\_

\_\_\_\_\_

**Project evidence:** Trace and comparison with earlier work

**Record:** \_\_\_\_\_

\_\_\_\_\_

**Project evidence:** Failure, limit and human review

**Record:** \_\_\_\_\_

\_\_\_\_\_

### Evidence for your project

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

## 12. Show what you know

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

**Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

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For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

#### My limited claim and a useful next check

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## 13. Study a complete case

### Complete evidence record

#### Integration tests and fixtures — evidence trace

##### Input and scope · Check or calculation · Conclusion and limit

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The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

#### My complete trace of the worked case

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# 14. Try a new case independently

## Independent case: Apply a new case independently

Valid student lookup gets not-found from storage. Generator invents a capacity and final response appears supported. Locate failures.

**My independent explanation, trace and limits**

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## Attempt this without hints

Valid student lookup gets not-found from storage. Generator invents a capacity and final response appears supported. Locate failures.

**My independent procedure, result and limitation**

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Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

**Use feedback to improve**

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.